

$$1. \int \frac{dx}{x^{\frac{2}{3}} + x^{\frac{1}{2}}}$$

$$= \int \frac{dx}{\sqrt[3]{x^2} + \sqrt{x}}$$

$$x = t^6$$

$$dx = 6t^5 dt$$

$$= \int \frac{6t^5 dt}{t^4 + t^3}$$

$$= \int \frac{6t^2 dt}{t+1}$$

$$= 6 \int \frac{t^2 - 1 + 1}{t+1} dt$$

$$= 6 \int (t-1 + \frac{1}{t+1}) dt$$

$$= 6 \left(\frac{1}{2} t^2 - t + \ln(t+1) \right) + C$$

$$= 3t^2 - 6t + 6 \ln(t+1) + C$$

$$x = t^{\frac{1}{6}} \quad = 3x^{\frac{1}{3}} - 6x^{\frac{1}{6}} + 6 \ln(x^{\frac{1}{6}} + 1) + C$$

$$2. \int \sqrt{1-x^2} dx$$

$$x = \cos t \quad dx = -\sin t dt$$

$$= \int \sqrt{1-\cos^2 t} (-\sin t) dt$$

$$= \int \sqrt{\sin^2 t} (-\sin t) dt$$

$$= \int -\sin^2 t \cdot dt$$

$$= - \int \sin^2 t dt$$

$$= - \int \frac{1 - \cos 2t}{2} dt$$

$$= -\frac{1}{2}t + \frac{1}{4}\sin 2t + C$$

$$t = \arccos x$$

$$x = \cos t$$

$$= -\frac{1}{2}\arccos x + \frac{1}{4}\sin 2\arccos x + C$$

$$\frac{1}{4}\sin 2t = \frac{1}{4} \cdot 2\sin t \cos t$$

$$= \frac{1}{2}\sqrt{1-x^2} \cdot x$$

$$(3) \int \frac{x}{\sqrt{1+x^2}} dx$$

$$t = x^2 \quad dt = 2x dx$$

$$= \int \frac{\frac{1}{2} dt}{\sqrt{1+t}}$$

$$= \frac{1}{2} \int (1+t)^{-\frac{1}{2}} dt$$

$$= \frac{1}{2} \cdot \left(\underline{2 \cdot (1+t)^{\frac{1}{2}}} \right) + C$$

$$= \sqrt{1+t} + C$$

$$= \boxed{\sqrt{1+x^2}} + C$$

验算

$$(\sqrt{1+x^2})' = \frac{1}{2\sqrt{1+x^2}} \cdot (1+x^2)'$$

$$= \frac{1}{2\sqrt{1+x^2}} (2x)$$

$$= \frac{x}{\sqrt{1+x^2}}$$

(4) $\int \frac{1}{x \sqrt{\ln x}} dx$ $t = \ln x$
 $\frac{1}{x} dx = dt$

$$\int \frac{1}{\sqrt{t}} dt$$

$$= 2\sqrt{t} + C$$

$$= 2\sqrt{\ln x} + C$$

(5) $\int \sqrt{1+x^2} dx$ ✓

$$\int \sqrt{x^2 + 4x + 5} dx$$

↓
消失

配方!

$$= \int \sqrt{(x+2)^2 + 1} dx$$

$t = x+2$

$$= \int \sqrt{t^2+1} dt$$

$$\boxed{t = \tan \theta}$$

$$t^2+1 = \frac{1}{\cos^2 \theta} = \sec^2 \theta$$

$$dt = (1 + \tan^2 \theta) d\theta$$

$$= (1 + t^2) d\theta$$

$$\Rightarrow \frac{dt}{dt} = (1 + \tan^2 \theta) d\theta$$

$$= \frac{1}{\cos^2 \theta} d\theta$$

$(1 + \tan^2 \theta)$.

$$= \int \frac{1}{\cos \theta} \cdot \frac{1}{\cos^2 \theta} d\theta$$

$$= \int \frac{1}{\cos^3 \theta} d\theta$$

$$= \int \frac{\cos \theta d\theta}{\cos^4 \theta \rightarrow (\cos^2 \theta)^2}$$

有理式

$$\boxed{u = \sin \theta}$$

$$= \int \frac{du}{(1-u^2)^2}$$

$$= \int \frac{du}{(1-u)^2 (1+u)^2}$$

$$= \frac{A}{(1-u)^2} + \frac{B}{1-u} + \frac{C}{(1+u)^2} + \frac{D}{1+u}$$

$$(1-u^2)(1-u)$$

$$= \frac{A(u+1)^2 + B(1-u)(u+1)^2 + C \cdot (u-1)^2 + D(1+u)(1-u)^2}{(1-u)^2(1+u)^2}$$

$$= \frac{A(u^2+2u+1) + B(-u^3-u^2+u+1) + C(u^2-2u+1) + D(u^2-u^2-u+1)}{(1-u)^2(1+u)^2}$$

$$= \frac{(D-B)u^3 + (A+C-B-D)u^2 + (2A-2C+B-D)u + (A+B+C+D)}{(1-u)^2(1+u)^2}$$

$$\begin{cases} D-B=0 & \textcircled{1} \\ A+C-B-D=0 & \textcircled{2} \\ 2A-2C+B-D=0 & \textcircled{3} \\ A+B+C+D=1 & \textcircled{4} \end{cases}$$

$$\Rightarrow \textcircled{1} \rightarrow \textcircled{2}, \textcircled{3}, \textcircled{4}$$

$$A + C - 2B = 0$$

$$A = C$$

$$A + C + 2B = 1$$

$$\Rightarrow A = C = B = D = \frac{1}{4}$$

$$\int \frac{du}{(1-u)^2(1+u)^2}$$

$$= \int \frac{1}{4} \frac{1}{(1-u)^2} + \frac{1}{4} \frac{1}{1-u} + \frac{1}{4} \frac{1}{(1+u)^2} + \frac{1}{4} \frac{1}{1+u} du$$

$$= \frac{1}{4} \left(\frac{1}{1-u} - \ln(1-u) - \frac{1}{1+u} \right)$$

$$+ \ln(1+u) + C$$

$$= \frac{1}{4} \left(\frac{1}{1 - \frac{x+2}{\sqrt{1+(x+2)^2}}} - \left[n \left(1 - \frac{x+2}{\sqrt{1+(x+2)^2}} \right) - \frac{1}{1 + \frac{x+2}{\sqrt{1+(x+2)^2}}} + \ln \left(1 + \frac{x+2}{\sqrt{1+(x+2)^2}} \right) \right] \right) + C$$

$$u = \sin \theta$$

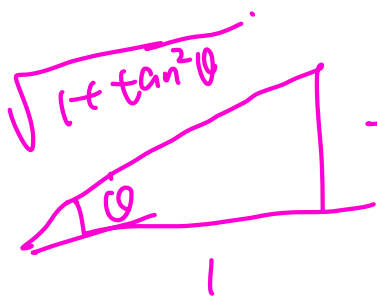
$$t = \tan \theta$$

$$t = x+2$$

$$x \rightarrow u ?$$

$$u = \sin \theta$$

$$t = \tan \theta$$



$$\begin{aligned} \tan \theta \quad u = \sin \theta &= \frac{\tan \theta}{\sqrt{1+\tan^2 \theta}} = \frac{t}{\sqrt{1+t^2}} \\ &= \frac{x+2}{\sqrt{(x+2)^2+1}} \end{aligned}$$

$$\int \sqrt{1+t^2} \, dt$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\Rightarrow \cosh^2 x - \sinh^2 x = 1$$

$$\cosh^2 x = 1 + \sinh^2 x$$

$$t = \frac{\sinh u}{\cosh u}$$

$$dt = \cosh u \, du$$

$$\int \sqrt{\cosh^2 u} \cdot (\cosh u \, du)$$

$$= \int \cosh^2 u \, du$$

$$\cosh^2 u = \frac{1 + \cosh 2u}{2}$$

$$= \int \frac{1 + \cosh 2u}{2} du$$

$$= \frac{1}{2} u + \frac{1}{4} \sinh 2u + C$$

$$= \frac{1}{2} u + \frac{1}{2} \sinh u \cosh u + C$$

$$\begin{aligned} t &= \sinh u \\ u &= \operatorname{arcsinh} t \end{aligned} = \frac{1}{2} \operatorname{arcsinh} t + \frac{1}{2} t \cdot \sqrt{1+t^2} + C$$

$$= \frac{1}{2} \operatorname{arcsinh}(x+2)$$

$$+ \frac{1}{2} (x+2) \sqrt{1+(x+2)^2} + C$$

2.

$$(1) \int_0^1 \underline{x \sqrt{1-x^2}} \underline{dx}$$

$$t = x^2 \quad dt = 2x dx.$$

$$x \in [0, 1]$$

$$t \in [0, 1]$$

$$\int_0^1 \frac{1}{2} (1-t)^{\frac{1}{2}} dt$$

$$= \left(\underbrace{-\frac{1}{3}}_{\uparrow} \underbrace{(1-t)^{\frac{3}{2}}}_{\uparrow} \right) \Big|_0^1$$

$$= \left(0 - \left(-\frac{1}{3}\right) \right)$$

$$= \frac{1}{3}$$

$$\left((1-t)^{\frac{3}{2}} \right)' = \frac{3}{2} (1-t)^{\frac{1}{2}} \cdot (-1)$$

$$(2) \int \ln x \, dx$$

$$= x \ln x - \int x \, d \ln x$$

$$= x \ln x - \int x \cdot \frac{1}{x} \, dx$$

$$= x \ln x - \int 1 \, dx$$

$$= \underline{x \ln x - x + C}$$

$$\int_1^e \ln x \, dx = (\ln x - x) \Big|_1^e$$

$$= 1$$

$$(3) \cdot \int_0^1 e^x \cos x \, dx$$

$$= \int_0^1 e^x \, d \sin x$$

$$= \left(e^x \sin x \right) \Big|_0^1 - \int_0^1 \sin x \, de^x$$

$$= \textcircled{0} - \int_0^1 e^x \, d(-\cos x)$$

$$= \textcircled{0} - \left(\left(e^x (-\cos x) \right) \Big|_0^1 - \int_0^1 -\cos x \, de^x \right)$$

$$= \textcircled{0} - \textcircled{0} - \int_0^1 \cos x \, e^x \, dx$$

$$\Rightarrow 2 \int_0^1 \cos x e^x dx = 0 - 0$$

$$(4) \int \ln(1+x^2) dx$$

$$= x \ln(1+x^2) - \int x d \ln(1+x^2)$$

$$= x \ln(1+x^2) - \int x \cdot \frac{1}{1+x^2} \cdot (2x) dx$$

$$= x \ln(1+x^2) - 2 \int \frac{x^2}{1+x^2} dx$$

$$= x \ln(1+x^2) - 2 \int \left(1 - \frac{1}{1+x^2}\right) dx$$

$$= x \ln(1+x^2) - 2x + 2 \arctan x + C$$

$$(5) \int \frac{\ln x}{x} dx$$

$$= \ln \ln x + C$$

$$(\ln \ln x) \Big|_1^e \times \frac{e^2}{e}$$

$$= \ln \ln e - \ln \ln 1$$

$$= \ln 1 - \ln 0 \quad x$$

3. c) $\int_{-2025}^{2025} \frac{1}{1+2^x} dx$

$$f(x) = \frac{1}{1+2^x}$$

$$\begin{aligned} \underline{f(x) + f(-x)} &= \frac{1}{1+2^x} + \frac{1}{1+2^{-x}} \\ &= \frac{1+2^{-x} + 1+2^x}{(1+2^x)(1+2^{-x})} = \frac{1+2^{-x} + 1+2^x}{1+2^{-x}+2^x+1} \\ &= \underline{\underline{1}} \end{aligned}$$

$$= 1 \times (2025 - (-2025))$$

$$\begin{array}{ccccccc} & & 100 & & & & \\ & & \underbrace{\hspace{1cm}} & & & & \\ S & = & 1 & + & 2 & + & \dots & + & 100 \\ & & + & & & & & & \\ S & = & 100 & & 99 & & 98 & \dots & 1 \end{array}$$

$$= 101$$

$$(2) \int_0^{\pi} x \sin x \, dx$$

$$x \int \sin x \, dx$$

$$= - \int_0^{\pi} x \cos x$$

$$= - \left((x \cos x) \Big|_0^{\pi} - \int_0^{\pi} \cos x \, dx \right)$$

$$= - \left((-\pi - 0) - (\sin x) \Big|_0^{\pi} \right)$$

$$= \pi + (0 - 0)$$

$$= \pi$$

$$f(x) = x \sin x$$

$$\begin{aligned} f(x) + f(\pi-x) &= x \sin x + (\pi-x) \sin(\pi-x) \\ &= x \sin x + (\pi-x) \sin x \\ &= \boxed{\pi \sin x} \end{aligned}$$

$$\begin{aligned} 2 \int_0^{\pi} f(x) dx &= \int_0^{\pi} f(x) dx + \int_0^{\pi} f(x) dx \\ &= \int_0^{\pi} f(x) dx + \int_0^{\pi} f(\pi-x) dx \end{aligned}$$

$$\begin{aligned} t &= \pi - x \\ dt &= -dx \\ &= \int_0^{\pi} (\pi \sin x) dx \\ &= 2\pi \end{aligned}$$

$$\int_0^{\pi} f(\pi-x) dx = \int_{\pi}^0 f(t) (-dt)$$

$$= - \int_{\pi}^0 f(t) dt$$

$$= \int_0^{\pi} f(t) dt$$

$$(3) \quad f(x) = \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} \quad \left[0, \frac{\pi}{2}\right]$$

$$f(x) + f\left(\frac{\pi}{2} - x\right)$$

$$= \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} + \frac{e^{\sin\left(\frac{\pi}{2} - x\right)}}{e^{\sin\left(\frac{\pi}{2} - x\right)} + e^{\cos\left(\frac{\pi}{2} - x\right)}}$$

$$= \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} + \frac{e^{\cos x}}{e^{\cos x} + e^{\sin x}}$$

$$= 1$$

$$2 \int_0^{\frac{\pi}{2}} f(x) dx = \int_0^{\frac{\pi}{2}} 1 dx$$

$$= 1 \times \frac{\pi}{2}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} f(x) dx = \frac{\pi}{4}$$

$$(4) \int_{-\infty}^{\infty} \frac{x^{2025}}{1+x^2} dx \quad \text{ts}$$

$$f(x) + f(-x) = 0$$

$$f(-x) = \frac{-x^{2025}}{1+x^2} = -f(x)$$

$$\Rightarrow f(x) + f(-x) = 0$$

$$2 \int_{-1}^1 f(x) dx = \int_{-1}^1 0 dx$$

$$\underline{\underline{= 0}}$$

$$(5) f(x) = \ln \frac{1+x}{1-x}$$

$$f(-x) = \ln \frac{1-x}{1+x}$$

$$f(x) + f(-x) = \ln \frac{1+x}{1-x} \cdot \frac{1-x}{1+x}$$

$$= \ln 1$$

$$\underline{\underline{= 0}}$$

c17. $I_n = \int_0^1 x^n e^x dx$

$\int_0^1 e^x \cdot \underbrace{x^n dx}_{\frac{1}{n+1} x^{n+1}}$

$$= \int_0^1 x^n \underline{e^x dx}$$

$$= \int_0^1 x^n de^x$$

$$= x^n e^x \Big|_0^1 - \int_0^1 e^x dx^n$$

$$= e - n \int_0^1 e^x \cdot x^{n-1} dx$$

$$= e - n I_{n-1}$$

$$I_0 = \int_0^1 e^x dx = e - 1$$

$$(2) \quad I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx \quad (n \geq 2)$$

$$X = \left(x \sin^n x \right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \underline{x \, n \sin^{n-1} x \cos x \, dx}$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x \, dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \, d(-\cos x)$$

$$= \left(-\sin^{n-1} x \cos x \right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x) d \sin^{n-1} x$$

$$= 0 + \int_0^{\frac{\pi}{2}} \cos x \, (n-1) \sin^{n-2} x \cos x \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx$$

$$= (n-1) \left(\int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - \int_0^{\frac{\pi}{2}} \sin^n x \, dx \right)$$

$$= (n-1) (I_{n-2} - I_n)$$

$$\Rightarrow I_n = (n-1) (I_{n-2} - I_n)$$

$$\Rightarrow I_n = \frac{n-1}{n} I_{n-2}$$

$$\underline{I_{2k}} = \frac{2k-1}{2k} I_{2k-2}$$

$$= \frac{2k-1}{2k} \cdot \frac{2k-3}{2k-2} I_{2k-4}$$

$$= \frac{(2k-1)(2k-3) \cdots \cdot 1}{(2k)(2k-2)(2k-4) \cdots \cdot 2} \cdot I_0$$

$$I_{2k+1} = \frac{(2k)(2k-2) \cdots \cdot 2}{(2k+1) \cdots \cdot 3} \cdot I_1$$

$$I_0 = \int_0^{\frac{\pi}{2}} \sin^0 x \, dx$$

$$= \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sin x \, dx$$

$$= (-\cos x) \Big|_0^{\frac{\pi}{2}} = 1$$

$$\{ \} \quad I_n = \int_0^{\frac{\pi}{2}} \tan^n x \, dx \quad (\text{m2y}) \quad (\tan x)' = \sec^2 x$$

$$= \int_0^{\frac{\pi}{2}} \tan^{n-2} x \cdot \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{2}} \tan^{n-2} x \cdot d(\tan x - 1)$$

$$= (\tan x - 1) \tan^{n-2} x \Big|_0^{\frac{\pi}{4}}$$

$$- \int_0^{\frac{\pi}{4}} (\tan x - 1) d \tan^{n-2} x$$

$$= 0 - \int_0^{\frac{\pi}{4}} (\tan x - 1) \cdot (n-2) \tan^{n-3} x \cdot (\frac{d \tan x}{dx}) dx$$

关于 $\tan x$ 的多项式

$$= - (n-2) \int_0^{\frac{\pi}{4}} \underline{(\tan x - 1) (\tan^2 x + 1) \tan^{n-3} x} dx$$

$$= - (n-2) \int_0^{\frac{\pi}{4}} (\tan^3 x - \tan^2 x + \tan x - 1) \tan^{n-3} x dx$$

$$= - (n-2) \int_0^{\frac{\pi}{4}} (\tan^n x - \tan^{n-1} x + \tan^{n-2} x - \tan^{n-3} x) dx$$

$$I_n = - (n-2) (\bar{I}_n - \bar{I}_{n-1} + \bar{I}_{n-2} - \bar{I}_{n-3})$$

$$\Rightarrow \bar{I}_n = \frac{-(n-2)}{n-1} (-\bar{I}_{n-1} + \bar{I}_{n-2} - \bar{I}_{n-3})$$

$$I_3 = \frac{-1}{2} (-I_2 + I_1 - I_0)$$

$$I_0 = \int_0^{\frac{\pi}{4}} 1 \, dx = \frac{\pi}{4}$$

$$I_1 = \int_0^{\frac{\pi}{4}} \tan x \, dx \quad (-\ln \cos x)'$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} \, dx = \tan x$$

$$= \int_0^{\frac{\pi}{4}} \frac{-d \cos x}{\cos x}$$

$$= \left(-\ln \cos x \right) \Big|_0^{\frac{\pi}{4}}$$

$$= \left(-\ln \frac{\sqrt{2}}{2} - (-\ln 1) \right)$$

$$= -\ln \frac{\sqrt{2}}{2} = \ln \sqrt{2} = \frac{1}{2} \ln 2$$

$$I_2 = \int_0^{\frac{\pi}{4}} \tan^2 x \, dx$$

$$= (\tan x - 1) \Big|_0^{\frac{\pi}{4}}$$

$$= 0 - (-1) = 1$$

证明2:

$$1. \int_a^b f(x) \, dx = \int_a^b \underline{f(g(t)) g'(t)} \, dt$$

$$\text{左} = \underline{F(b) - F(a)}$$

$$\begin{aligned} \left(\underline{F(g(t))} \right)' &= F'(g(t)) \cdot g'(t) \quad F' = f \\ &= f(g(t)) g'(t) \end{aligned}$$

$$\text{右} = F(g(\beta)) - F(g(\alpha))$$

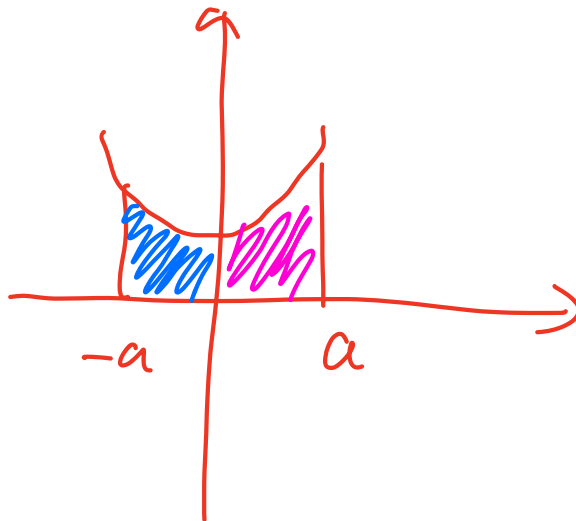
$$= \underline{F(b) - F(a)}$$

$$2. (u(x) v(x))' = u(x) v'(x) + u'(x) v(x)$$

$$\Rightarrow \int_a^b (u(x) v'(x) + u'(x) v(x)) dx$$

$$= (u(x) v(x)) \Big|_a^b$$

3.



$$I_2 = 0 + 0$$

$$I_0 = 20 \quad \begin{matrix} x & -a \rightarrow 0 \\ t & a \rightarrow 0 \end{matrix}$$

$$\int_{-a}^0 f(x) dx \quad \underline{\underline{t = -x}} \quad \int_a^0 f(-t) d(-t)$$

$$= - \int_a^0 f(-t) dt$$

$$f(-t) = f(t) \quad = \int_0^a f(-t) dt$$

$$= \int_0^a f(t) \, dt$$